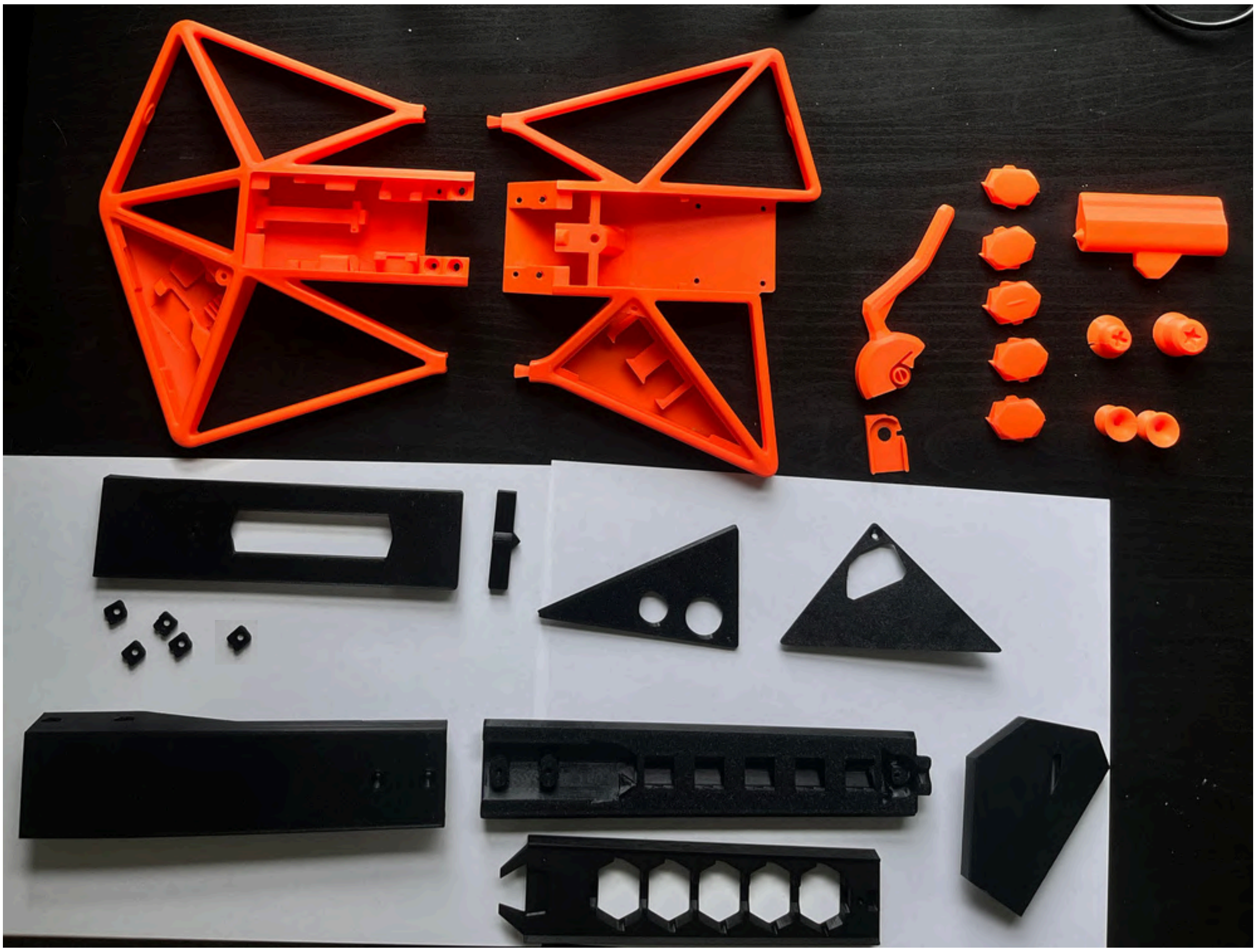
SHREDDER
BUILD GUIDE

A public domain guitar rhythm game controller

Github: https://github.com/joshuaua/SHREDDER

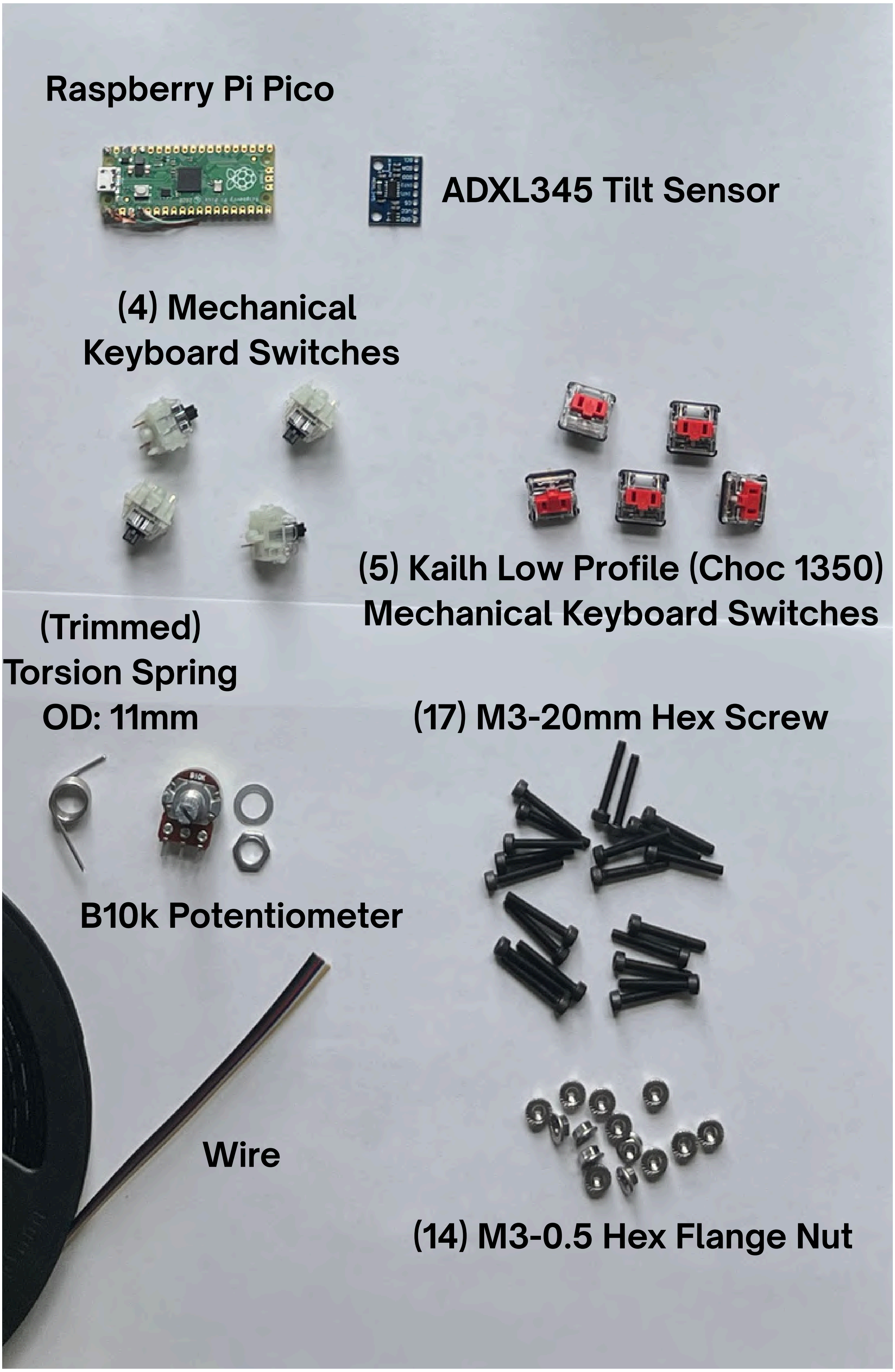
3D PRINTED PARTS

https://www.printables.com/model/1828635-shredder-a-public-domain-guitar-rhythm-game-contro



-No support Material
-No need to rotate models
-0.20mm Layer Height
-3 outer layers, 3 top layers, 3 bottom layers

Hardware & Electronics



Full Parts List

TOTAL COST PER UNIT: $40 - $50
TOTAL SHOPPING CART COST: $100 - $120

3D printed part cost estimates are based on $20/kg

Part	Quantity	Cost
BodyA	1	$2.64 (132 grams)
BodyB	1	$2.27 (114 grams)
Fret	4	$0.17 (8.5 grams)
Fret Line	1	$0.04 (2.2 grams)
Strumbar	1	$0.21 (10.7 grams)
Whammy	1	$0.16 (7.8 grams)
Start	1	$0.04 (1.8 grams)
Star Power	1	$0.06 (3.0 grams)
Strap Pin	2	$0.07 (3.5 grams)
Potentiometer House	1	$0.02 (1.0 gram)
Flange Nut Slide	5	$0.02 (1.0 gram)
Strum Insert	1	$0.04 (1.8 grams)
Strum Cover	1	$0.37 (18.4 grams)
Whammy Cover	1	$0.13 (6.3 grams)
Start Cover	1	$0.13 (6.3 grams)
Neck	1	$1.24 (62.0 grams)
Neck Top	1	$0.49 (24.5 grams)
Neck Bottom	1	$0.77 (38.5 grams)
Headstock	1	$0.43 (21.6 grams)
Raspberry Pi Pico	1	$4 (PiShop.us) $8 (Amazon)
Kailh Low Profile switch (1350 Choc)	5	$4.25 ($8.50 Amazon 10pk)
Cherry MX Black (Linear, Heavy)	4	$2.78 ($16 Amazon 23pk)
ADXL345 Tilt Sensor	1	$3.33 ($10 Amazon 3pk)
B10k Potentiometer	1	$0.70 ($7 Amazon 10pk)
Torsion Spring (11mm Dia. 50mm Len)	1	$0.75 ($7.5 Amazon 10pk)
24AWG 6 Color Wire	1	Around $2 (<3ft used of 32ft for $16)
M3-0.5 Hex Flange Nut	14	$4 ($5 Amazon 20pk)
M3-20mm Hex Screw	17	$2.26 ($8 Amazon 60pk)
Micro USB Cable	1	$7 (Amazon 1pk)

Step 1: Attach bodies together

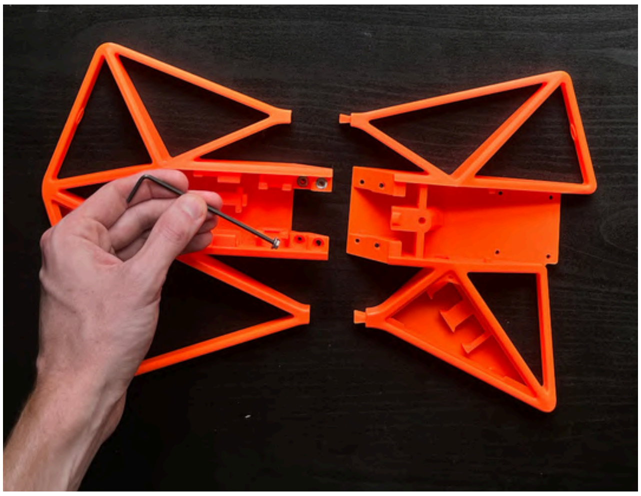
Parts Used

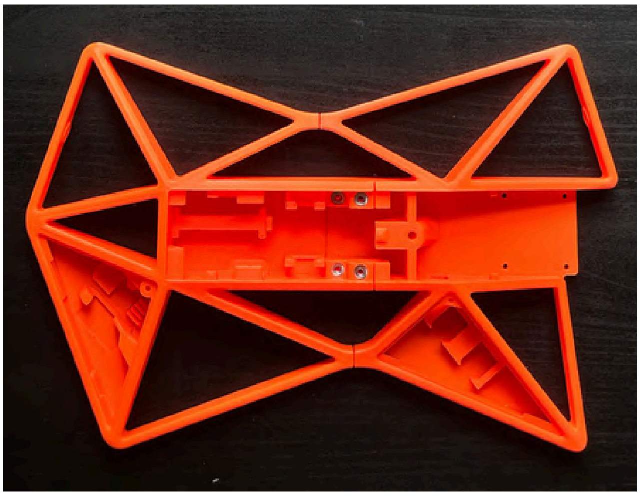
- BodyA
- BodyB
- (4) M3x20mm Hex Screw
- (4) M3 Hex Flange nut

Tools

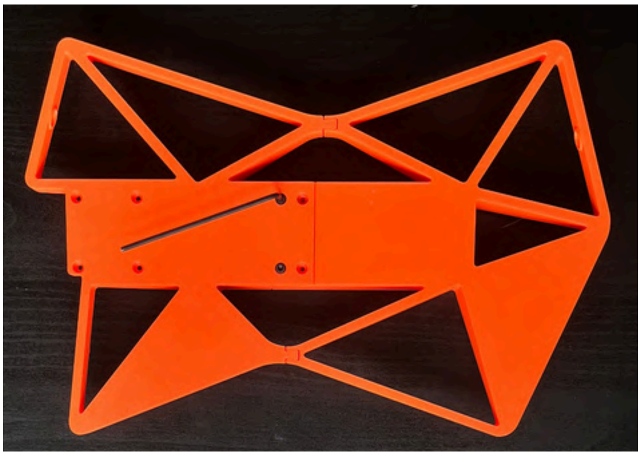
- 2.5mm Hex wrench

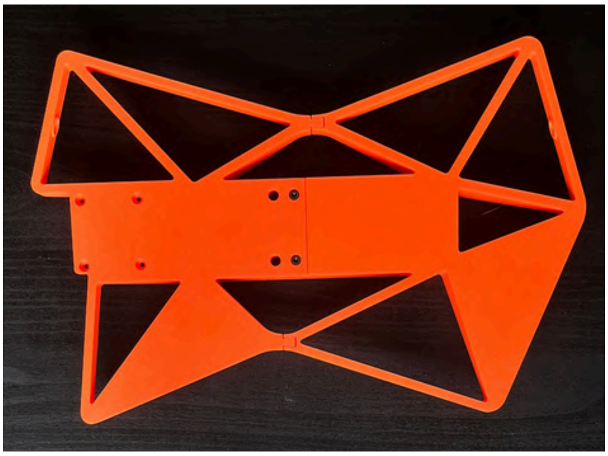
Slide bodies together and press 4 flange nuts into place

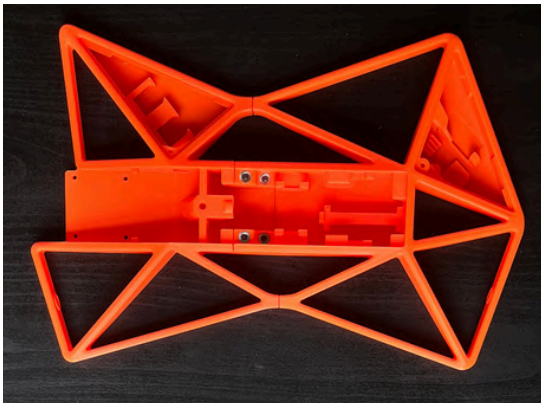




Flip over and secure 4 screws







Done

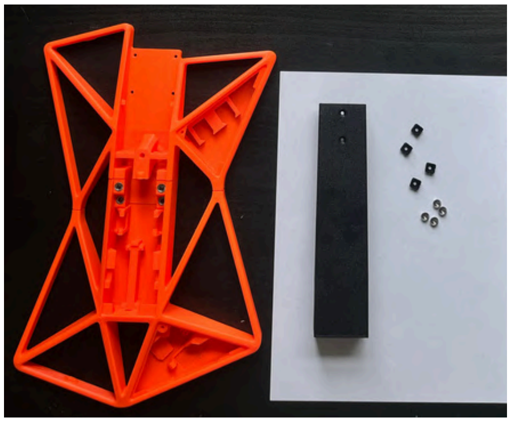
Step 2: Attach neck to body

Parts Used

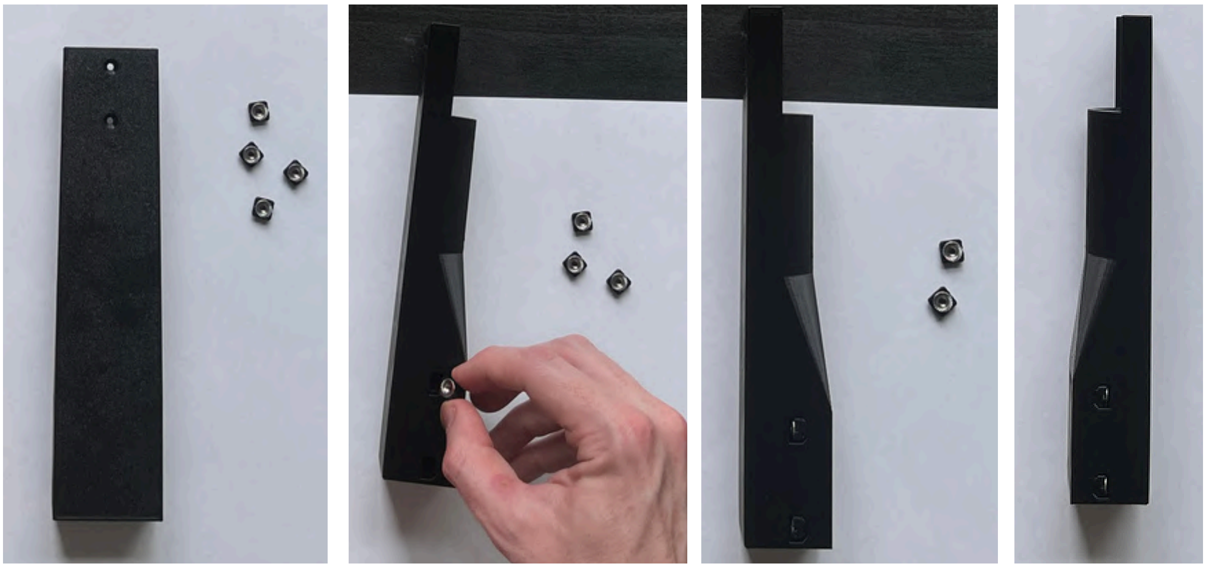
- Neck
- (4) Flange Nut Slide
- (4) M3x20mm Hex Screw
- (4) M3 Hex Flange nut

Tools

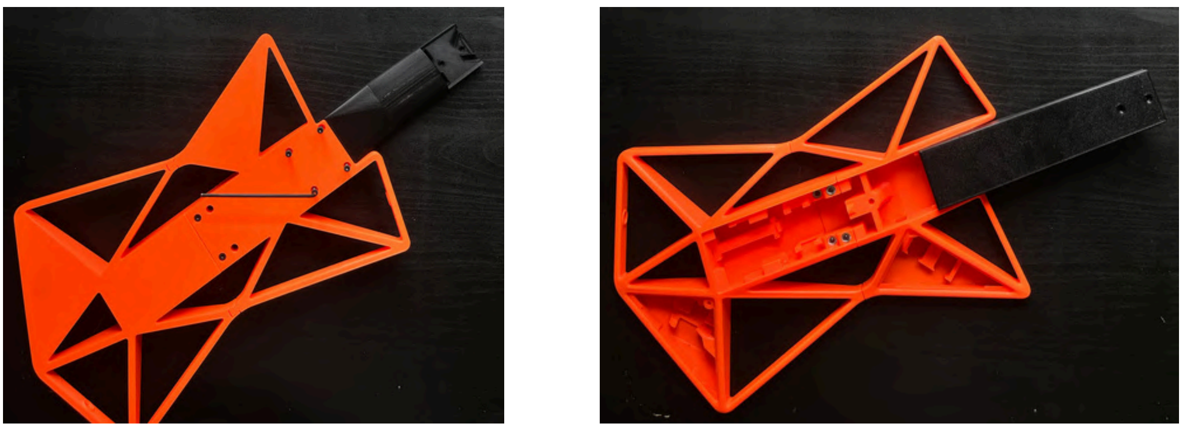
- 2.5mm Hex wrench



Press flange nuts into slides & insert into neck



Screw neck into body



Done

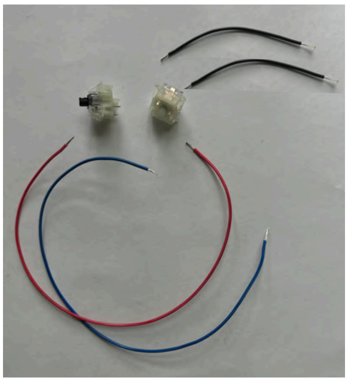
Step 3: Wire start/star power

Parts Used

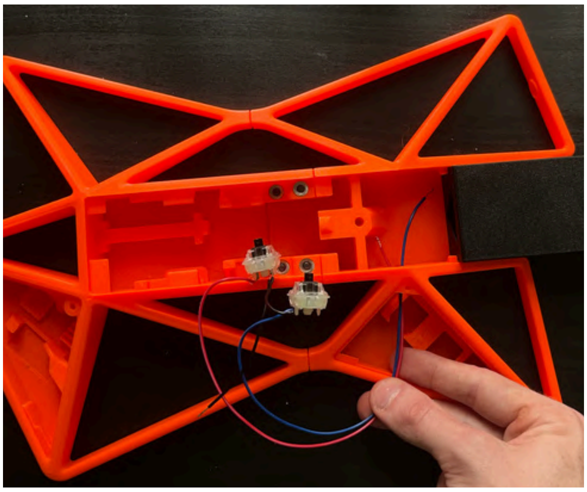
- (2) Mech switches
- Ground Wires: (2) 6cm
- Data Wires: (2) 20cm

Tools

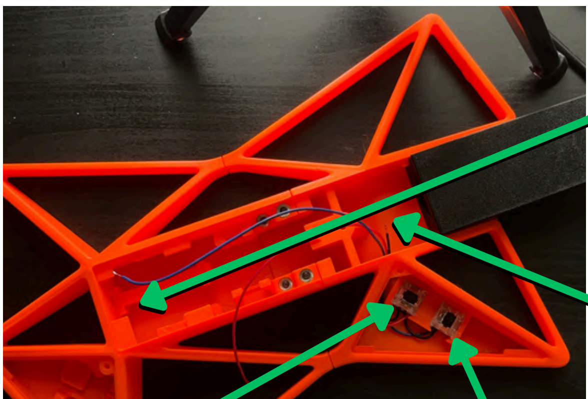
- Soldering Iron, solder
- wire stripper
- (Optional) solder flux
- (Optional) Hot Glue Gun




Solder 1 data wire & 1 ground wire to each switch



insert wires through body & snap switches in place



Ensure data wires can reach the Raspberry pi pico here

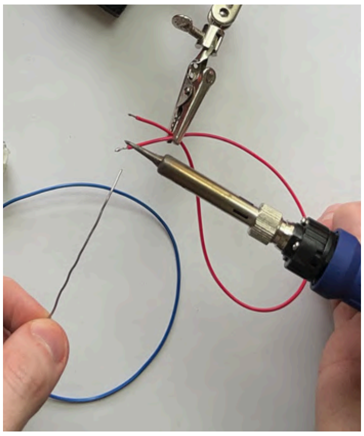
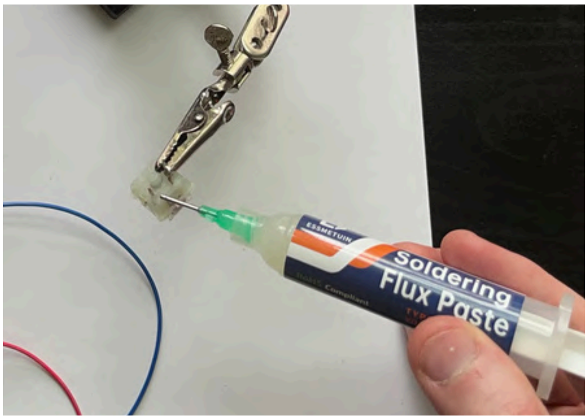
All digital switch ground wires will eventually be chained together here

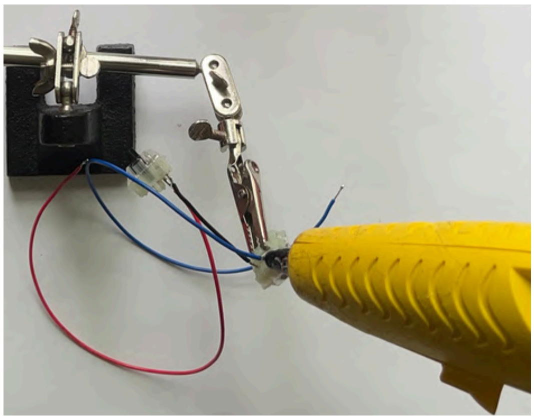
“Star Power” “Start”

Step 3 Done

PRO GAMER TIPS

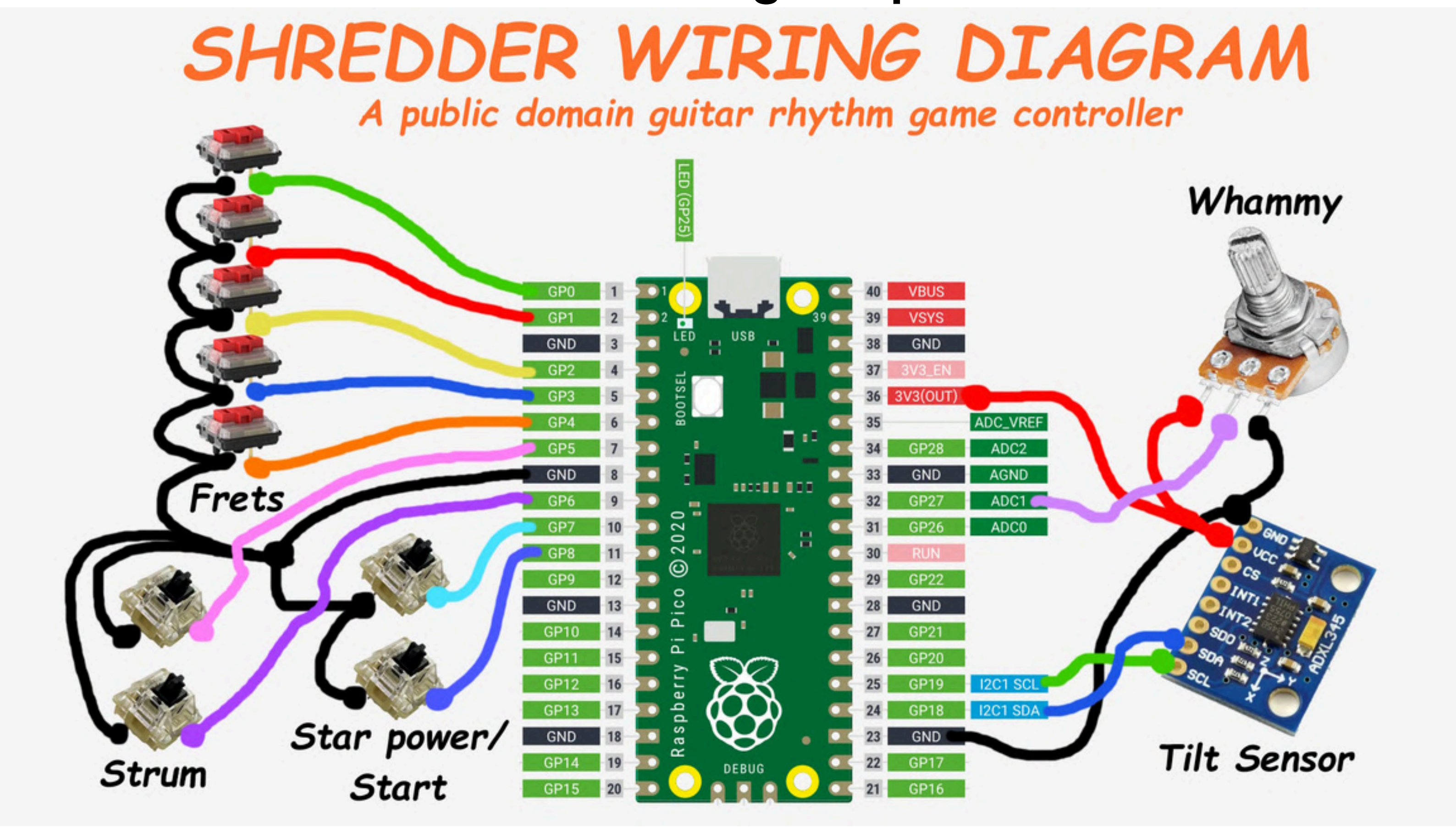
- Tin each wire with a little solder & trim to size
- Apply a tiny amount of flux to the switch pins
- Apply hot glue to strengthen solder joints



I waited until all electrical components were wired before soldering to the Raspberry Pi pico, but here's the wiring diagram for your reference:

- The ground connection for all digital switches will be connected to pin 8 on the Rpi pico.
- The "star power" data wire can go to pin 10 (GP7)
- The "start" data wire can go to pin 11 (GP8)



Step 4: Wire Strum up/down

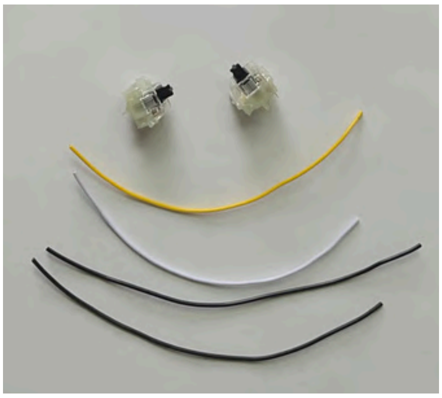
Parts Used

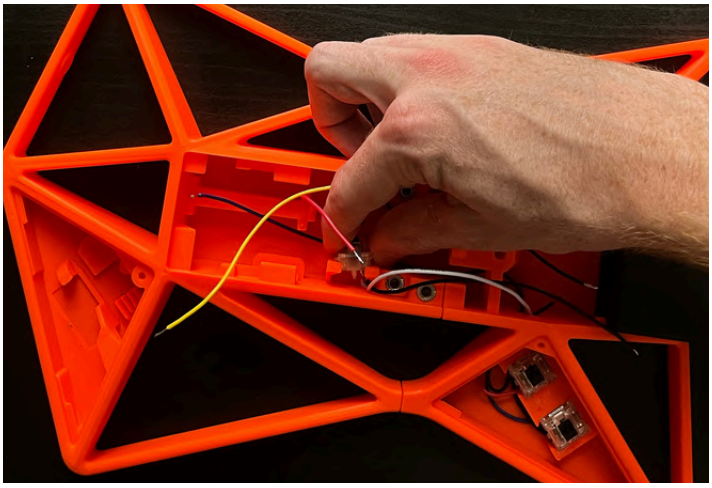
- (2) Mech switches
- Ground Wires: (2) 12cm
- Data Wires: (2) 10cm

Tools

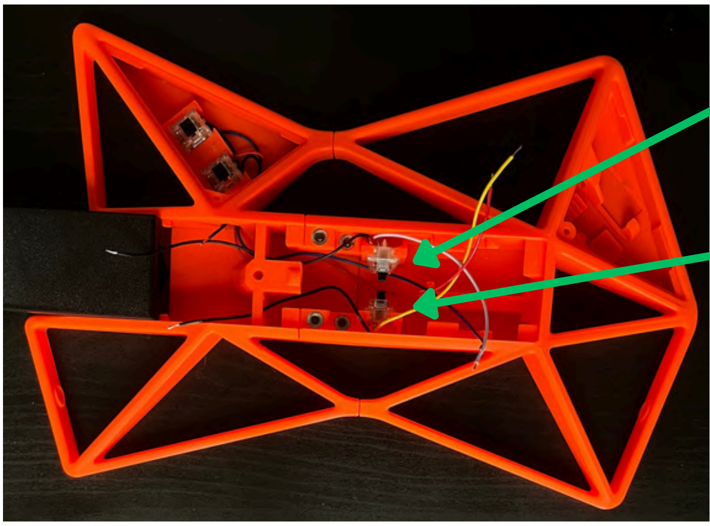
- Soldering Iron, solder
- wire stripper
- (Optional) solder flux
- (Optional) Hot Glue Gun

Solder 1 data wire & 1 ground wire to each switch





Snap switches in place



Wiring Reference:

“Strum up” data wire to pin 9 (GP6)

“Strum down” data wire to pin 7 (GP5)

Step 4 Done

Step 5: Wire potentiometer

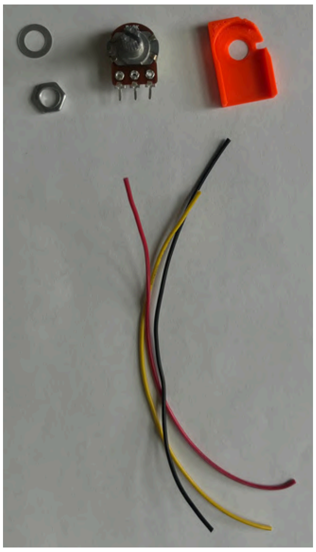
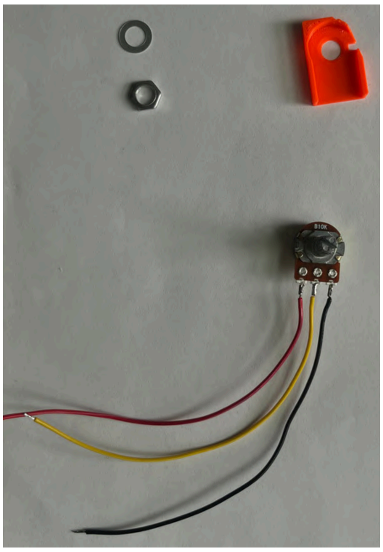
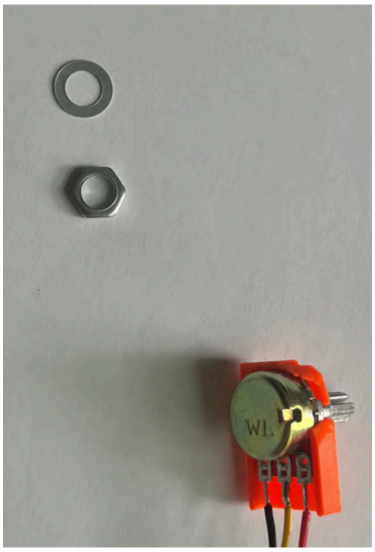
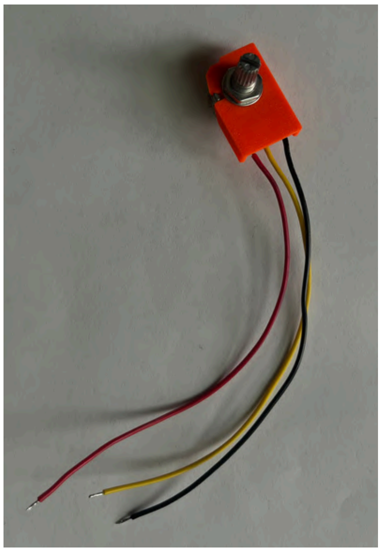
Parts Used

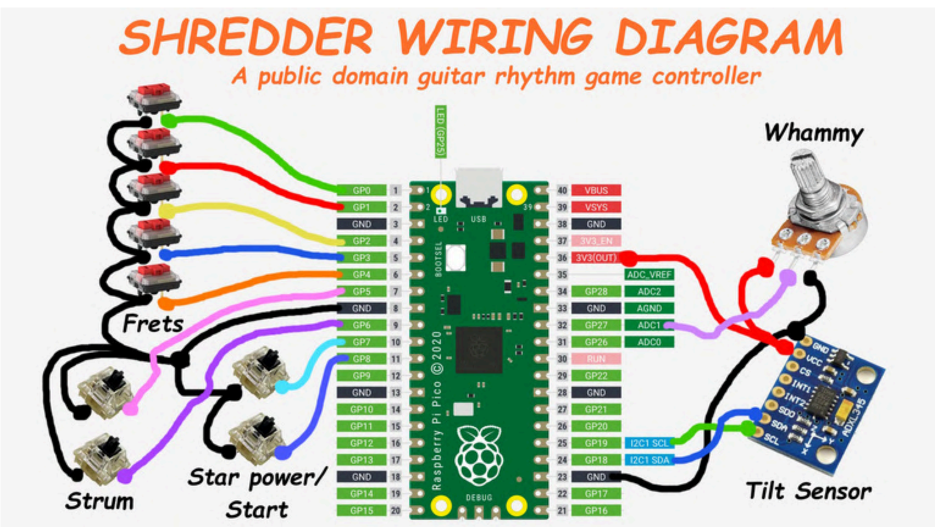
- B10k Potentiometer
- Potentiometer housing
- Power Wire: 10cm
- Ground Wire: 10cm
- Data Wire: 15cm

Tools

- Soldering Iron, solder
- wire stripper
- (Optional) solder flux
- (Optional) Hot Glue Gun

Solder 1 data wire (middle), 1 ground & 1 power wire and assemble potentiometer to housing



Wiring Reference:

- Data to pin 32 (GP27, ADC1)
- Power chained with VCC on tilt sensor & to pin 36 (3V3 out)
- Ground chained with GND on tilt sensor & to pin 23

Step 5 Done

Step 6: Assemble Whammy

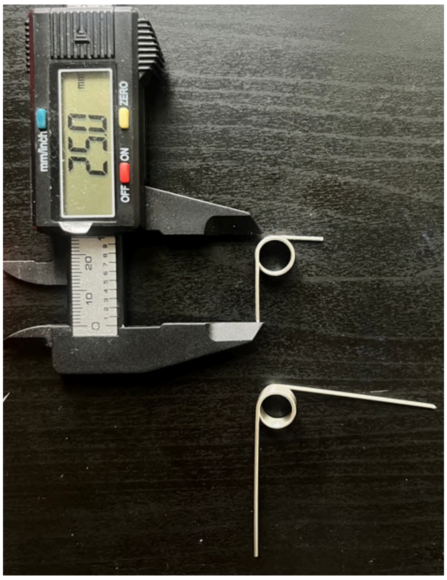
Parts Used

- Whammy
- Potentiometer w/ housing
- Torsion Spring

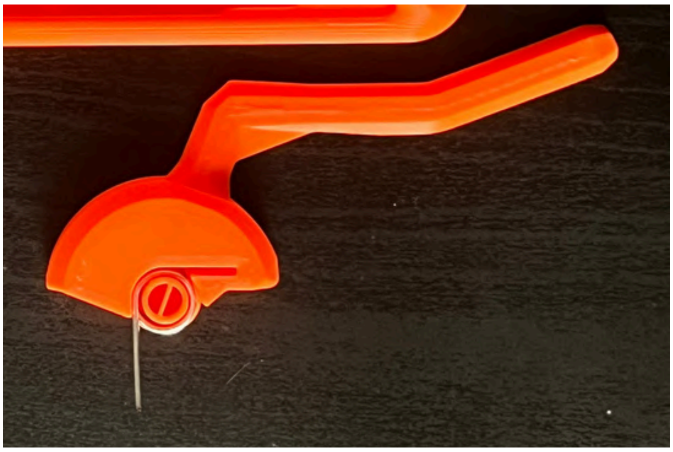
Tools

- Shear flush cutters, Hacksaw, or some way to trim the 1mm thick steel spring (I made it work with pliers)

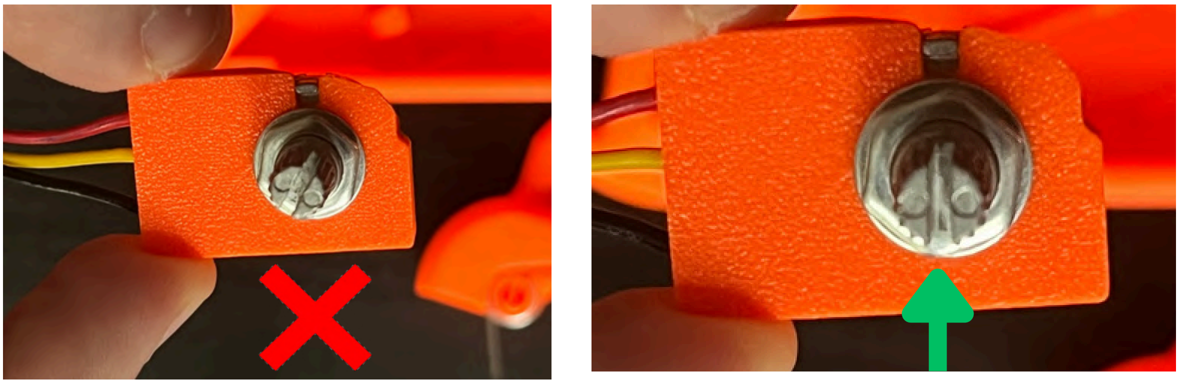
Trim the torsion spring to 20mm & 25mm as shown



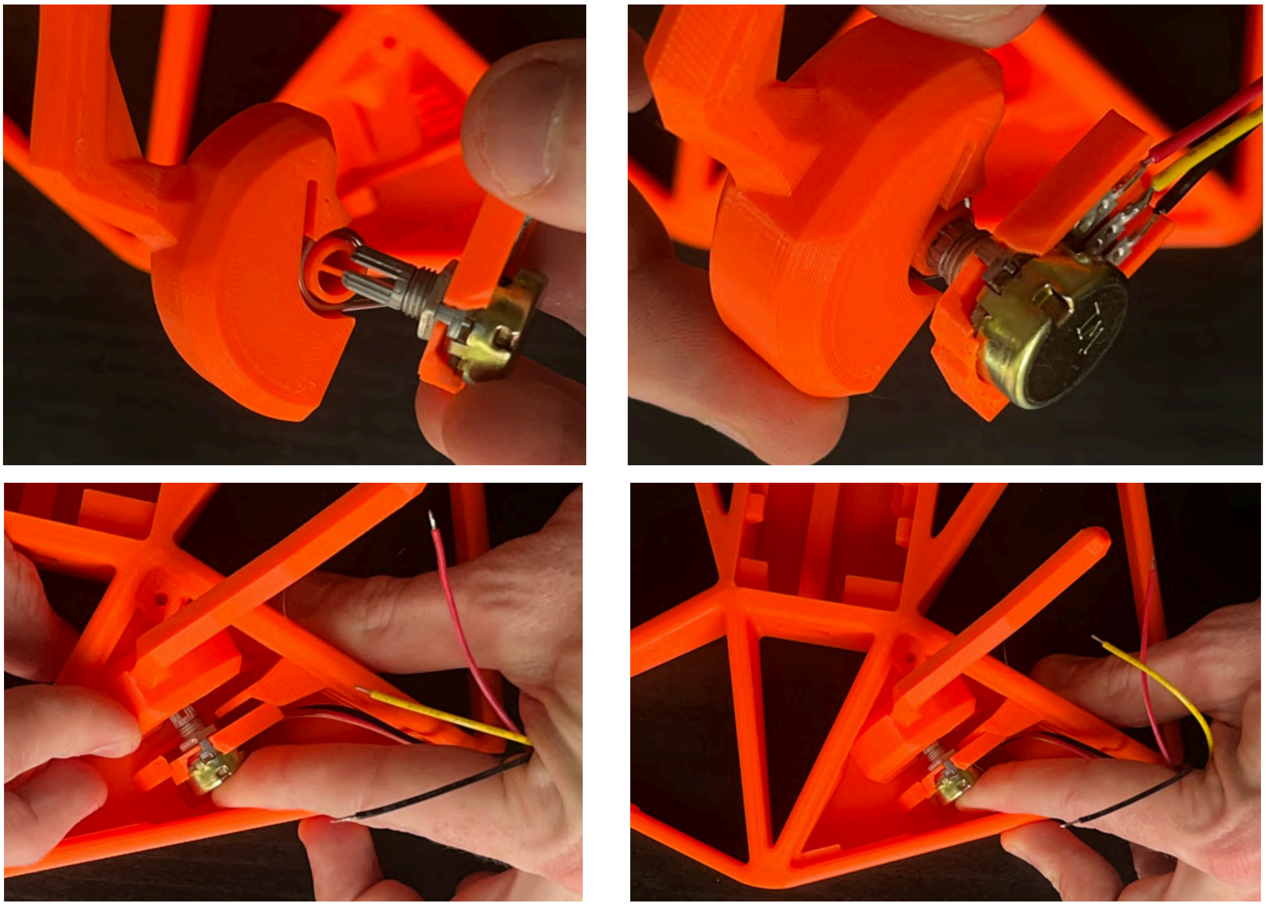
Insert the spring into the whammy bar, with the longer end sticking out



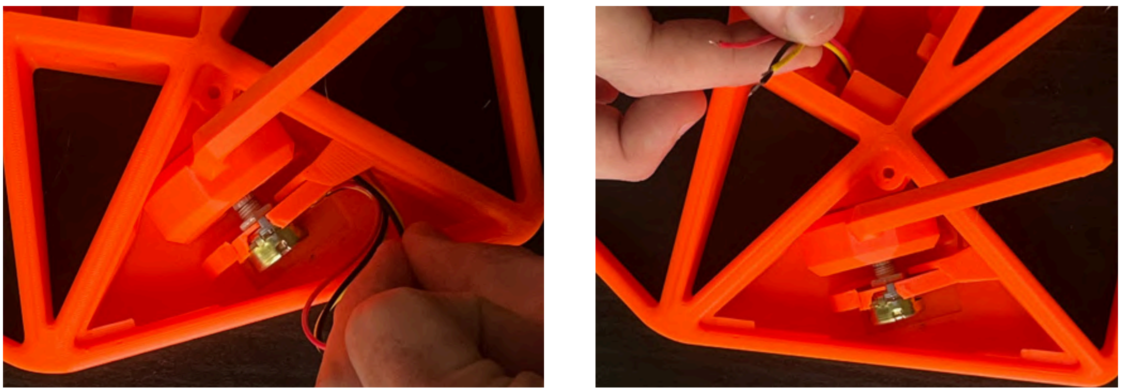
Align the potentiometer vertically as shown



Insert potentiometer and slide whammy assembly into place



Push wires through channel



Step 6 Done

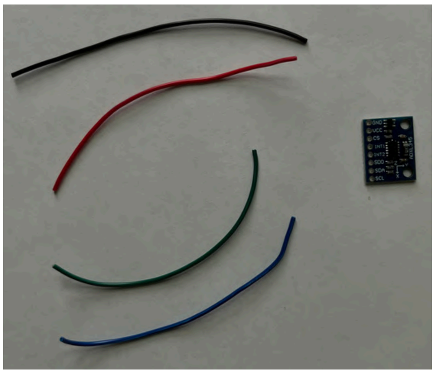
Step 7: Wire ADXL345 tilt sensor

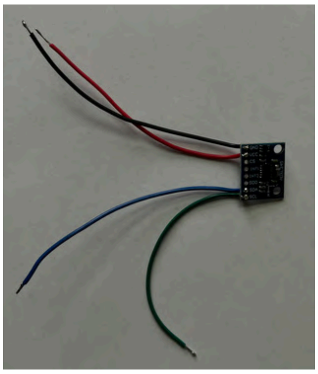
Parts Used

- ADXL345
- (4) Wires: 5cm

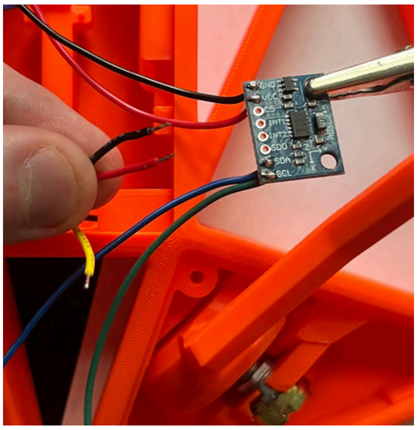
Tools

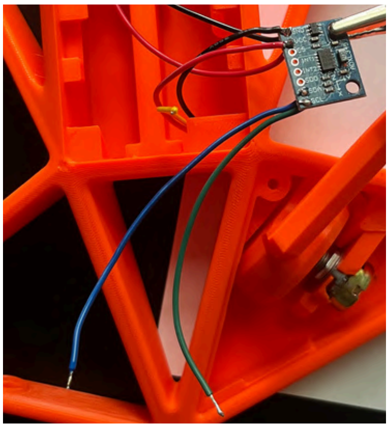
- Soldering Iron, solder
- wire stripper
- (Optional) Hot Glue Gun





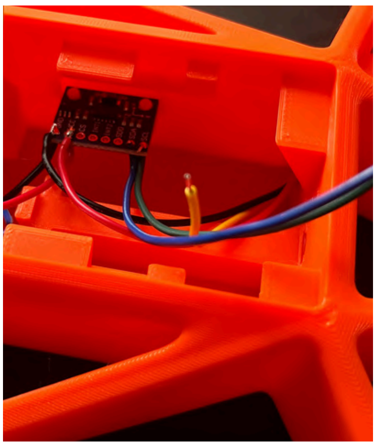
Solder wires to VCC, GND, SDA, and SCL

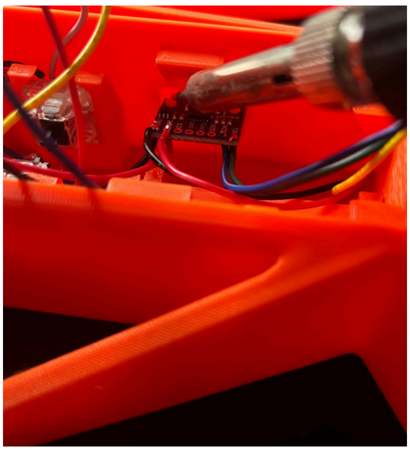


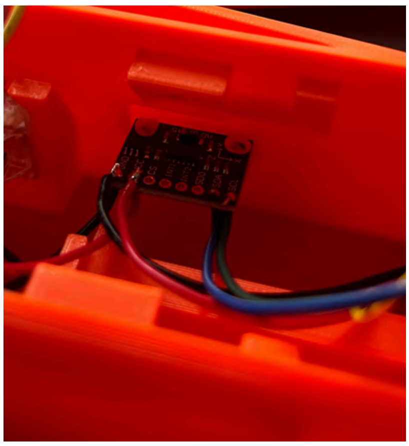


Solder Power and Ground wires from potentiometer to VCC and GND on the tilt sensor

To secure tilt sensor, use hot glue gun or Heat stake using soldering iron







Step 7 Done

Step 8: Wire & assemble frets

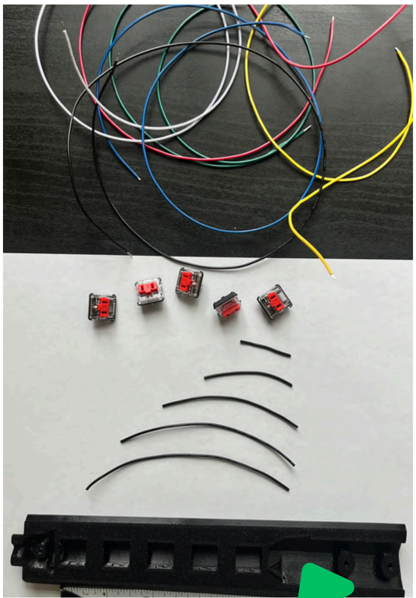
Parts Used

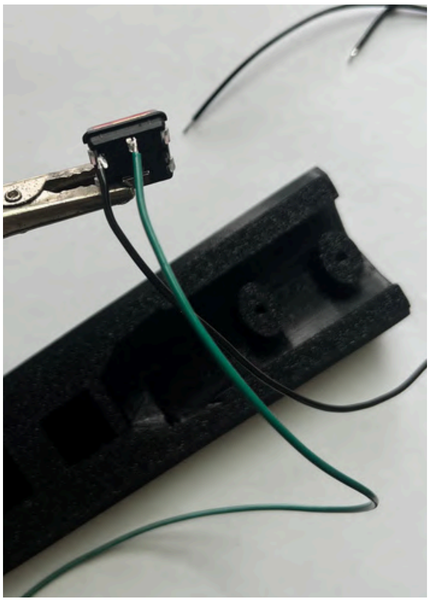
- Neck Bottom
- Flange nut slide
- (4) M3 Flange nuts & Screws
- (5) Low Profile Switch
- (5) data wires: 55cm
- (6) Ground Wires: 28cm, 12cm, 10cm, 8cm, 5cm, 3cm
- (Optional) strip board or similar

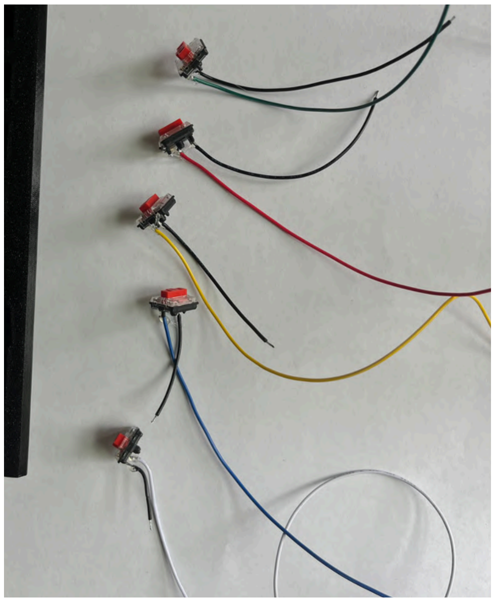
Tools

- Soldering Iron, solder
- wire stripper
- (Optional) Hot Glue Gun
- 2.5mm Hex wrench

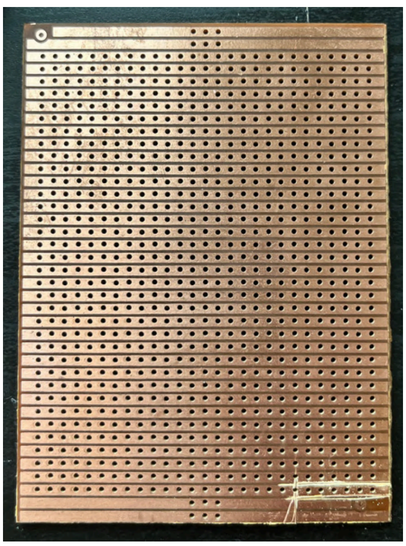
Solder a data and ground wire to each low profile switch

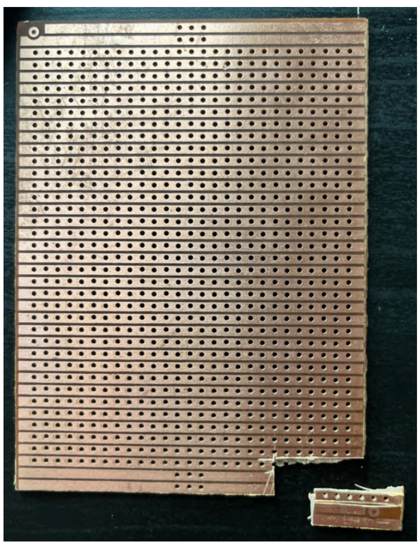






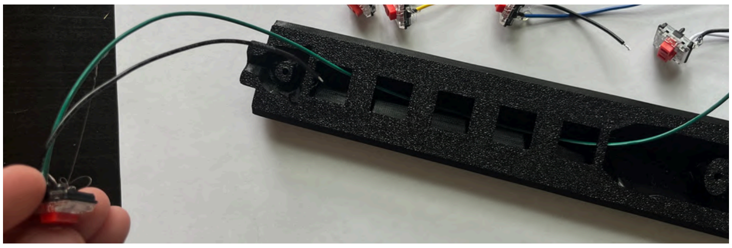
(5) shorter ground wires will be chained together here





Use any method you prefer to chain wires together. I used a small section of strip board

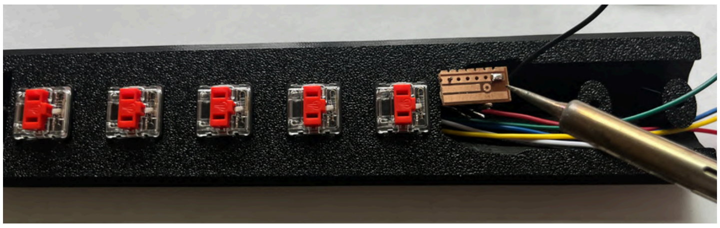
Slide wires through “Neck Bottom” and snap switch




Repeat for remaining switches



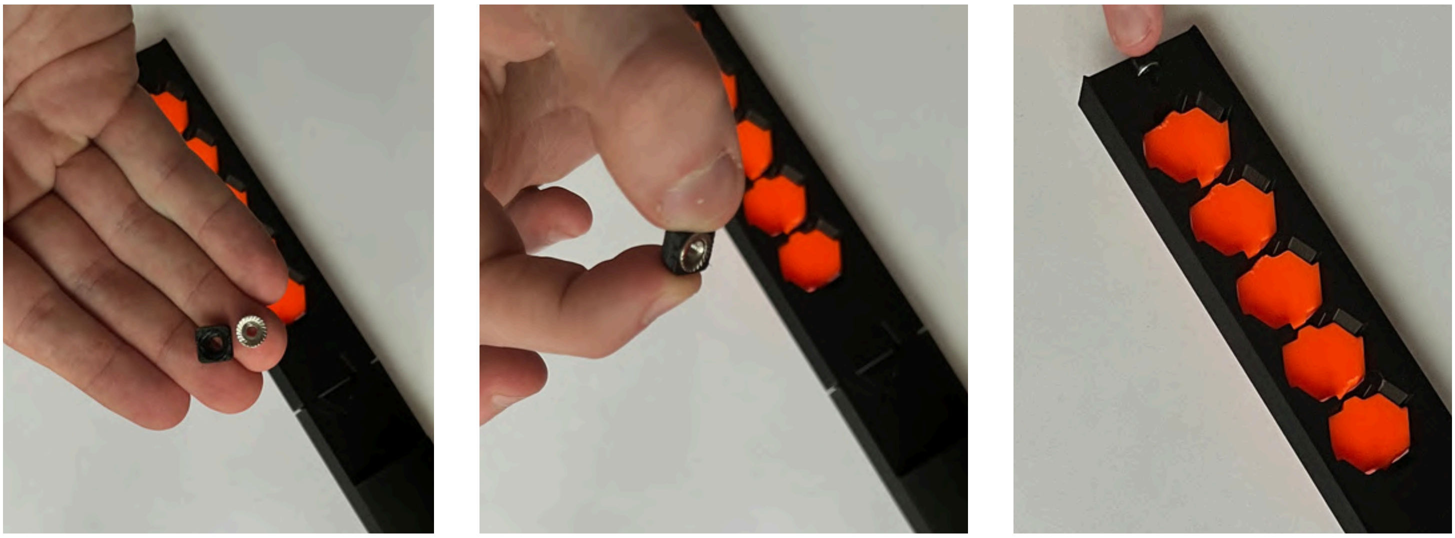

Chain grounds together and attach 28cm ground wire



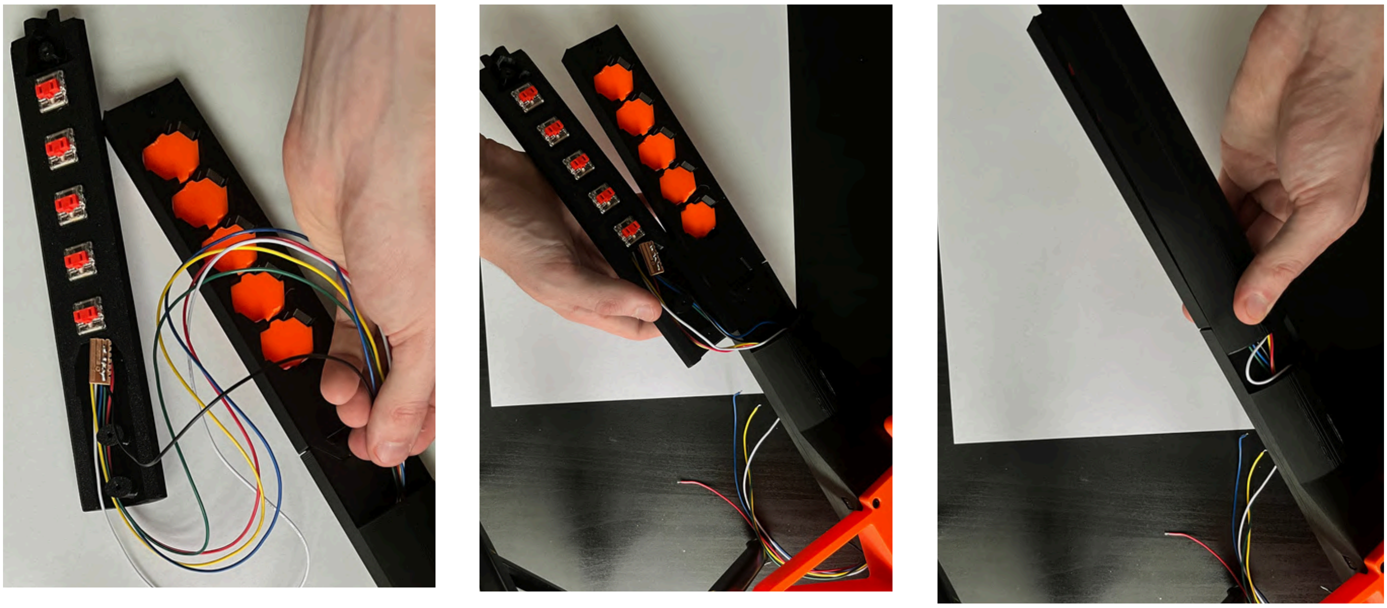


Snap “Neck Top” onto “Neck” and place frets

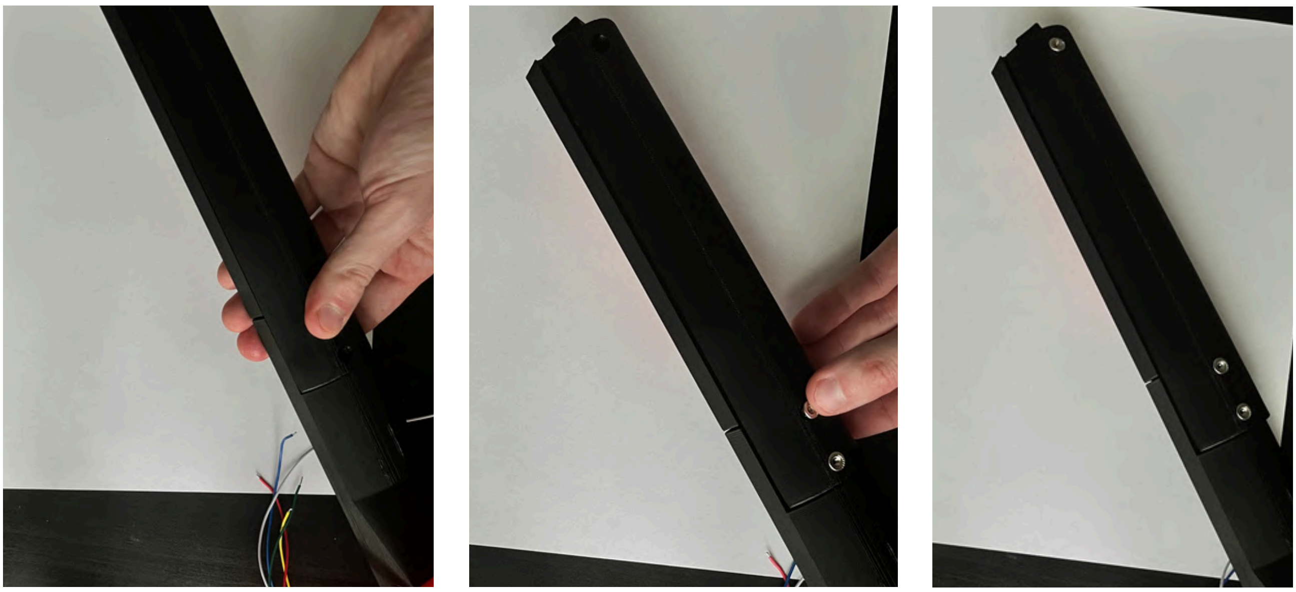
insert flange nut into “Flange Nut Slide” and place into “Neck Top”



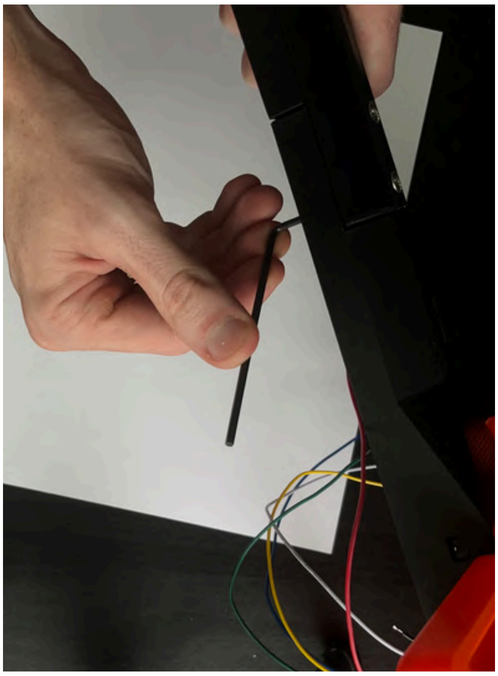
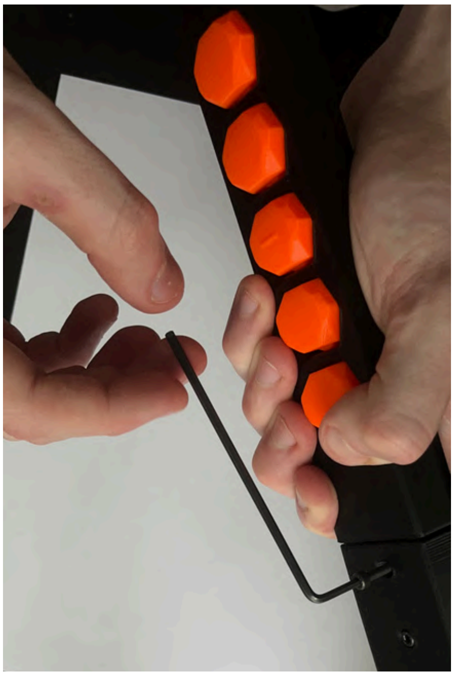
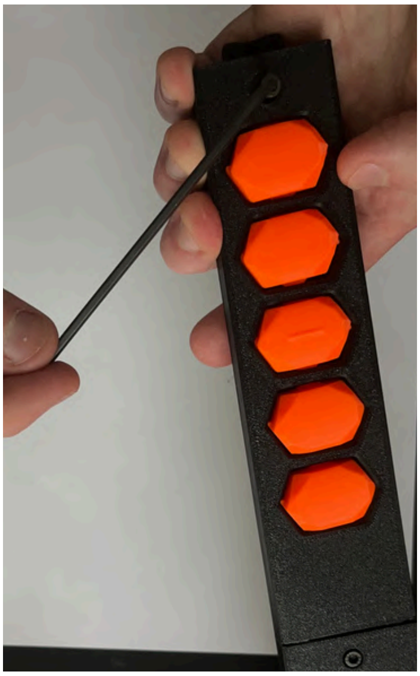
Slide wires through neck



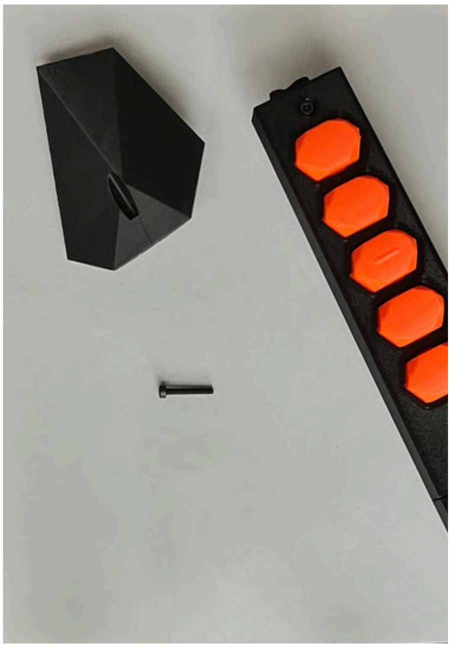
Fit “Neck Bottom” into place and insert (3) flange nuts



Hold neck together and secure (3) screws

Secure “headstock” with screw



Step 8 Done

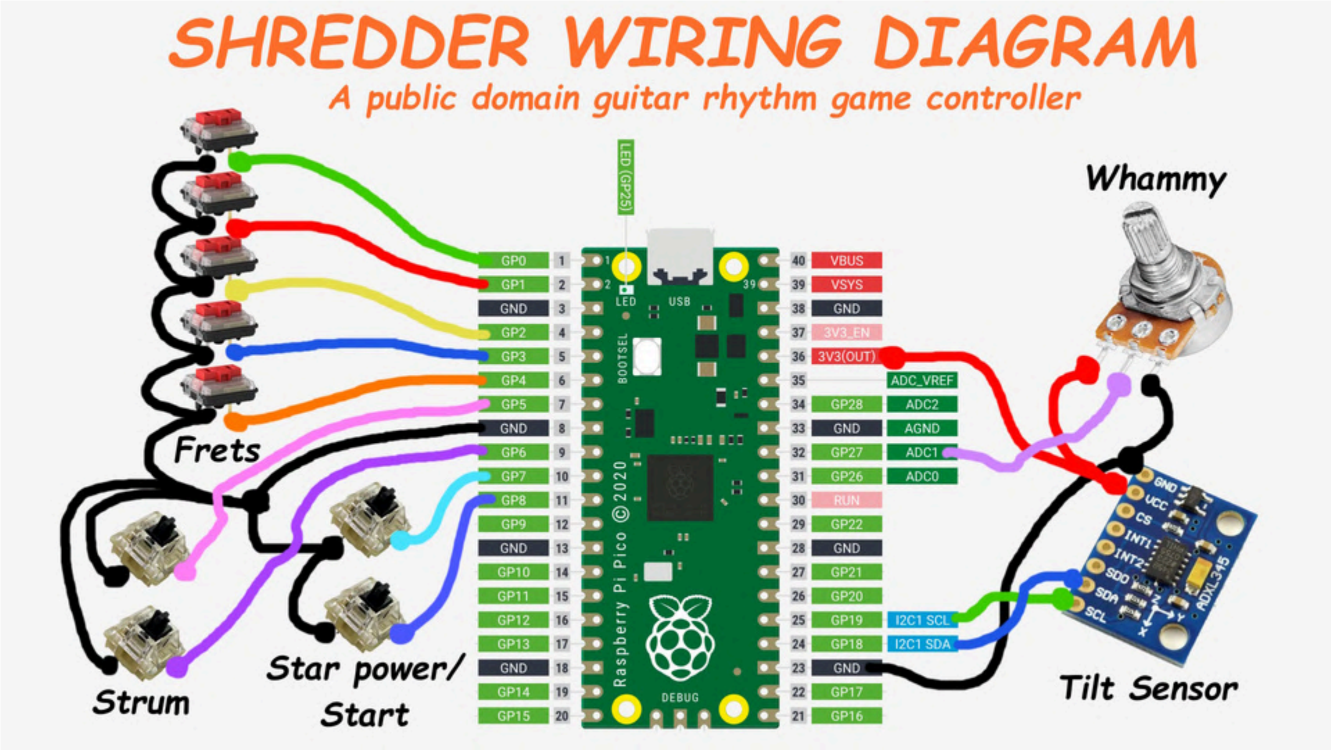
Step 9: Wire Raspberry Pi Pico

Parts Used

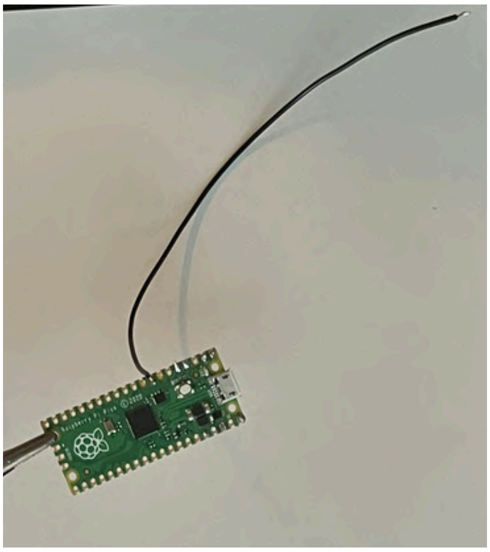
- Raspberry Pi Pico
- Ground wire: 18cm
- (Optional) strip board or similar

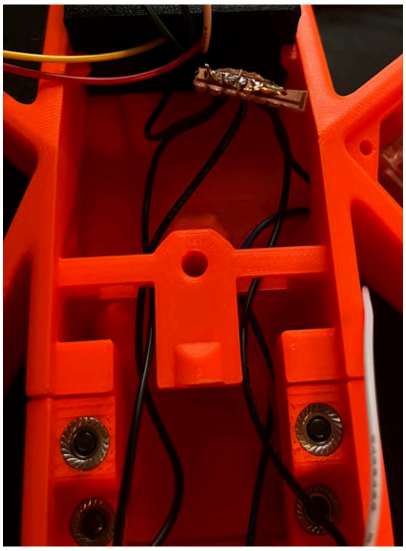
Tools

- Soldering Iron, solder
- wire stripper
- (Optional) Hot Glue Gun



Solder ground wire to pin 8 and chain all digital switch ground wires together

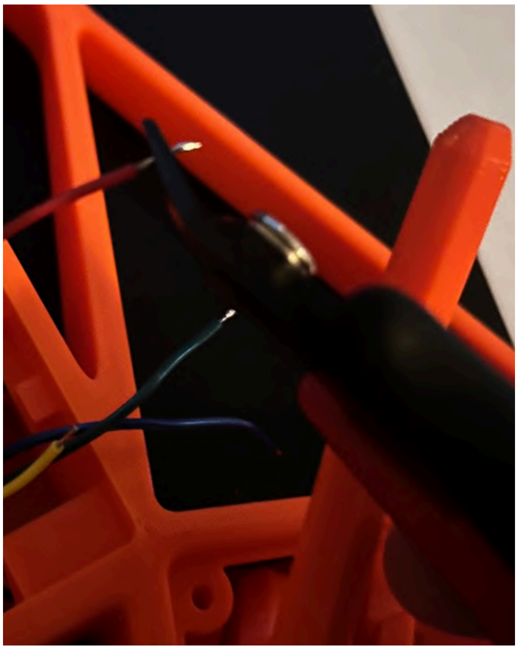


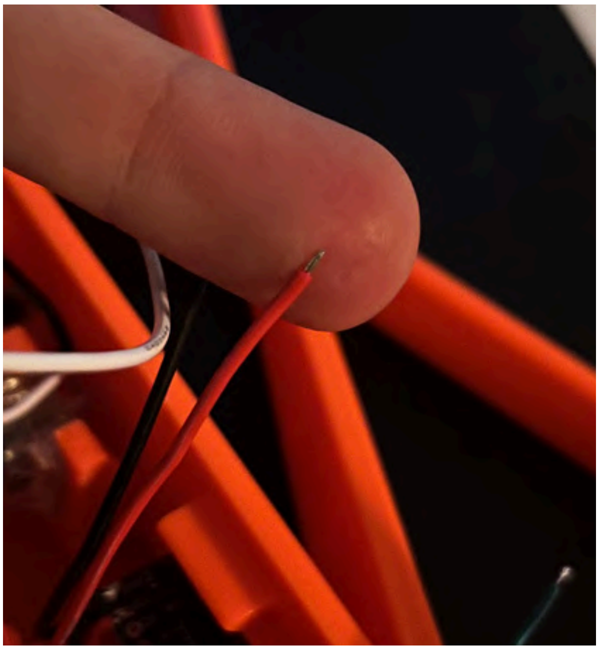


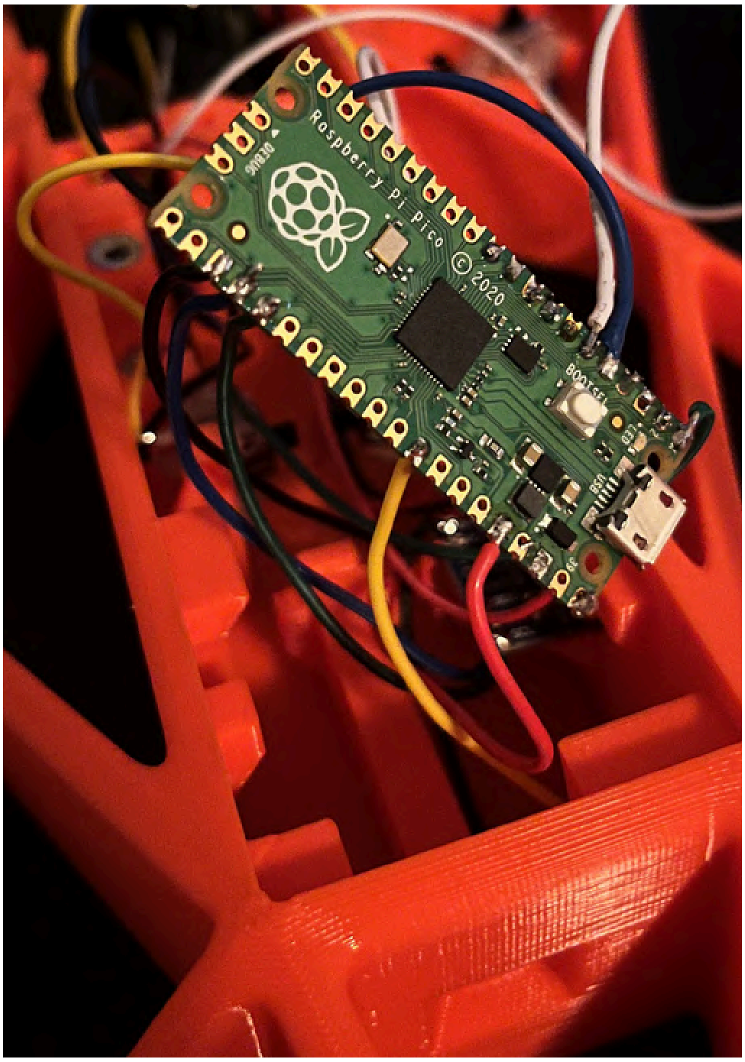
PRO GAMER TIPS

Before Soldering wires to Raspberry Pi Pico:

- **Tin each wire with a little solder**
- **Trim to only expose a tiny amount**







Finish Soldering

Step 9 Done

Step 10: Program & test electronics

Parts Used

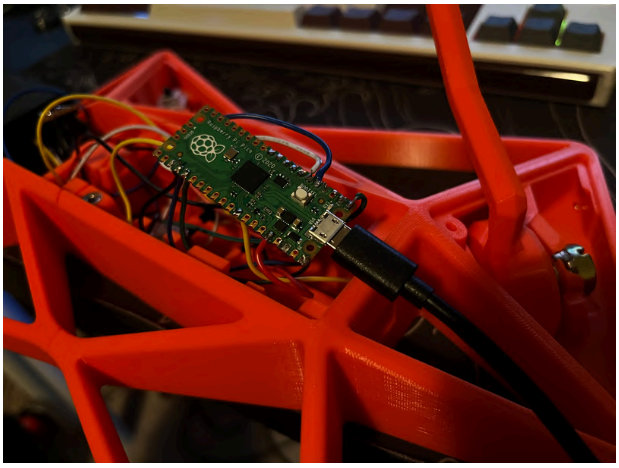
- Micro USB Cable

Tools

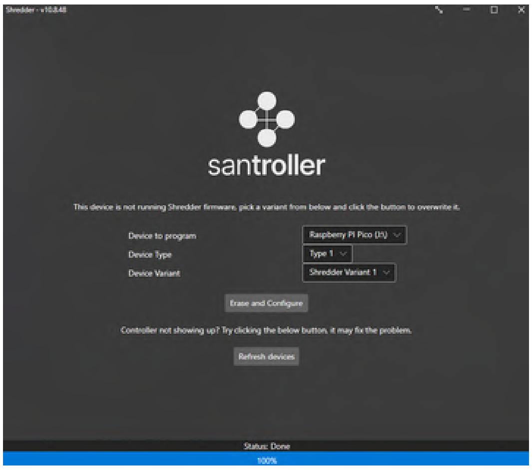
- Santroller Application (on github)

Github: https://github.com/joshuaua/SHREDDER

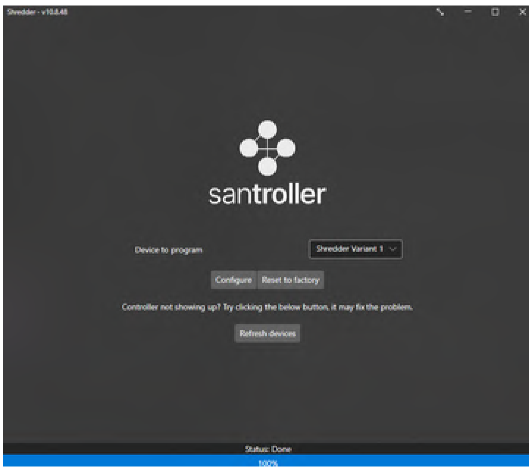
Note: this app allows for use on Windows, Linux, and MacOS. If you wish to learn how to play on console, please refer to the Santroller guide as I have not tried this: https://santroller.com/console_guides/



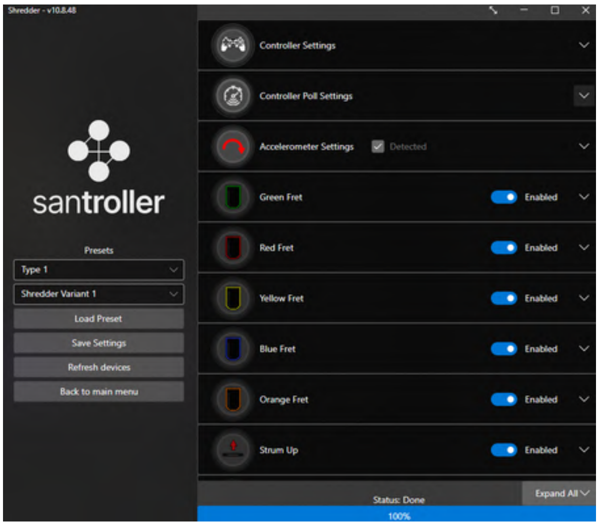
Plug in Raspberry Pi Pico. If it already has a code flashed on it, you can hold down the BOOTSEL button while plugging it in to reset it.



Click “Erase and Configure”



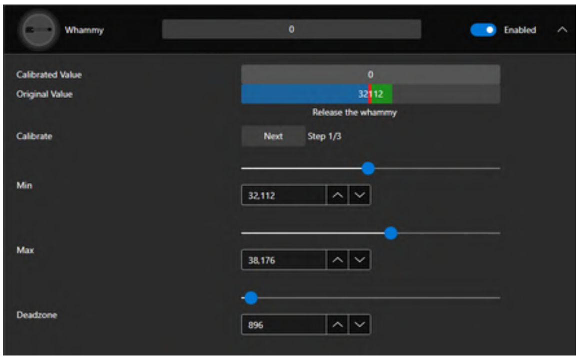
Click “Configure”



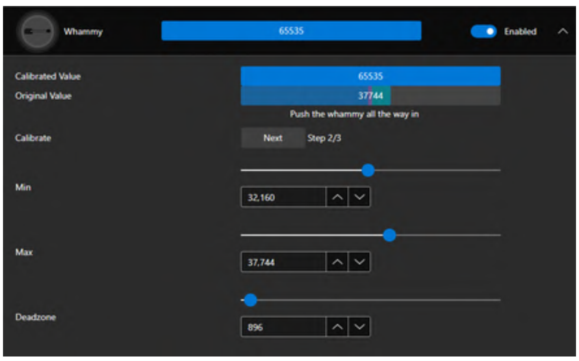
Press each digital button and verify that the corresponding icons light up

CALIBRATE WHAMMY

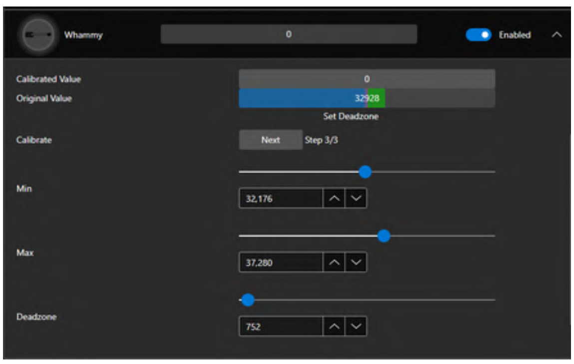
Click “Calibrate”



Click “Next”
to set minimum



Press whammy bar down
<u>fully</u> and click “Next”
to set maximum



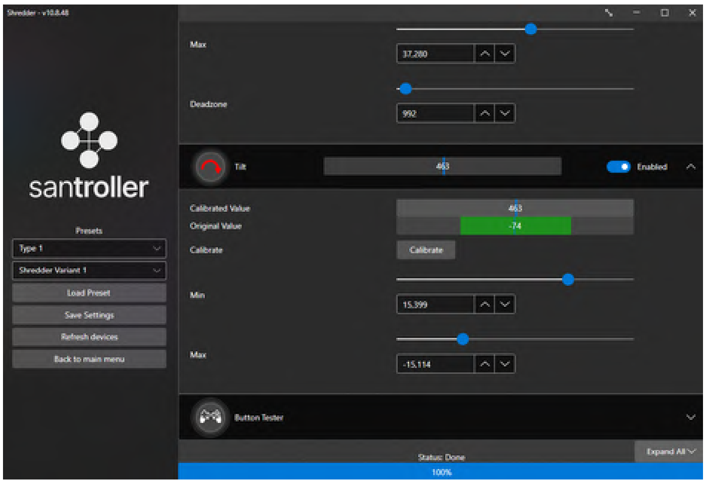
Press whammy bar down
<u>slightly</u> and click “Next”
to set deadzone

CALIBRATE TILT SENSOR

Click “Calibrate”

Hold controller horizontal and click “Next”

Hold controller as vertical as you would like and click “Next”



Click “Save Changes” and wait for the “Status: Done” to show at the bottom before closing the program.

Step 10 Done

Step 11: Assemble strumbar, star power/ start, and cover panels

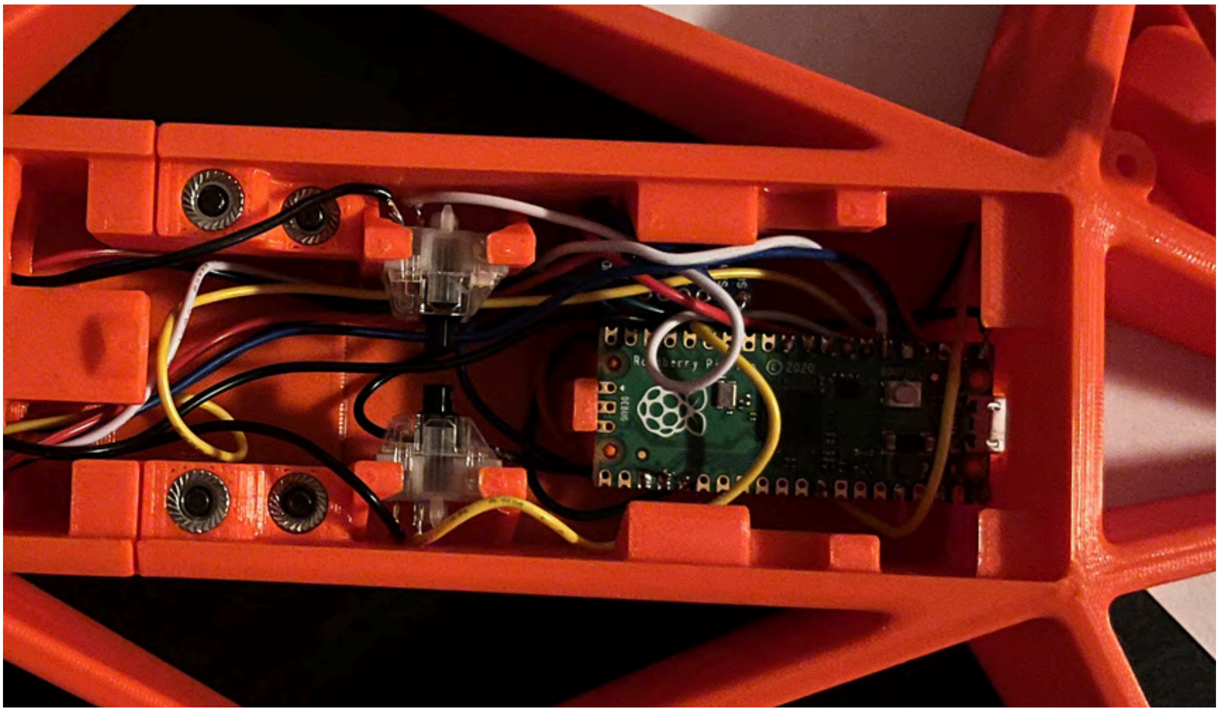
Parts Used

- Strumbar
- Strum Insert
- Start
- Select
- Strum Cover
- Start Cover
- Whammy Cover
- (3) M3 Screws

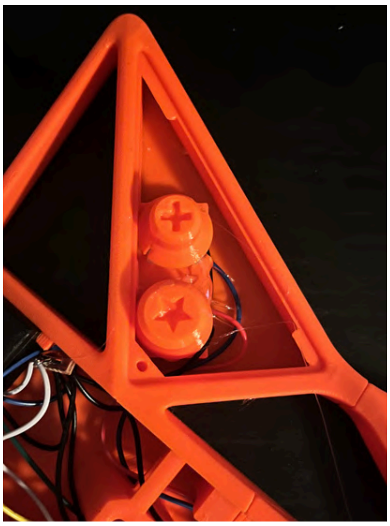
Tools

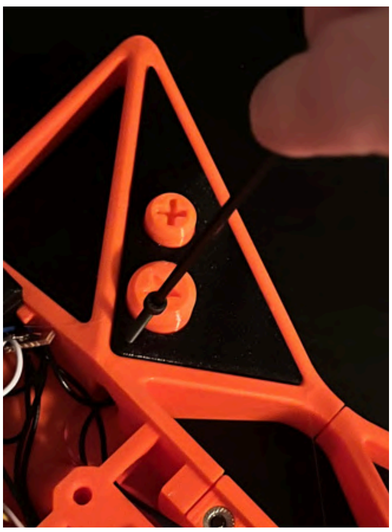
- 2.5mm Hex wrench
- (Optional) Hot Glue Gun

Unplug Raspberry pi pico and snap into place.



(I have since added more guides to contain the wires. Please try to do a nicer job managing the cables than this.)

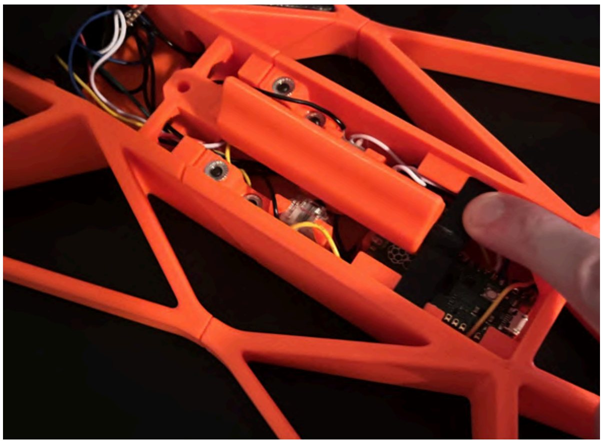




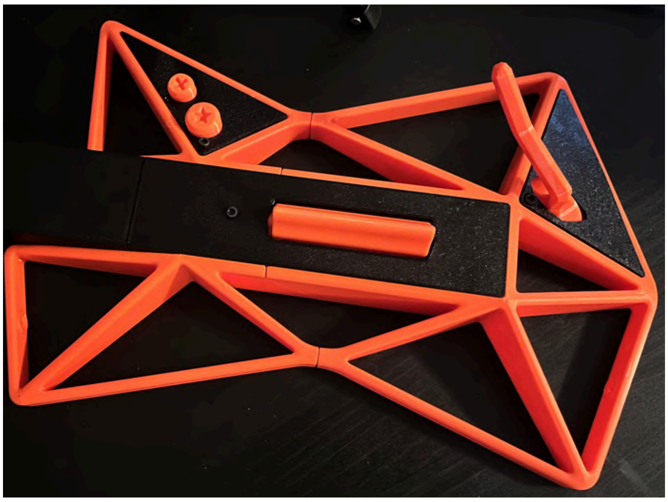
Place “start” and “star power” on switches and screw “start panel” on

Slide “Whammy Cover” through and screw on

Press “strum” with ”strum Insert” into place



Screw “strum cover” on



Step 11 Done

Step 12: Attach Strap Pins

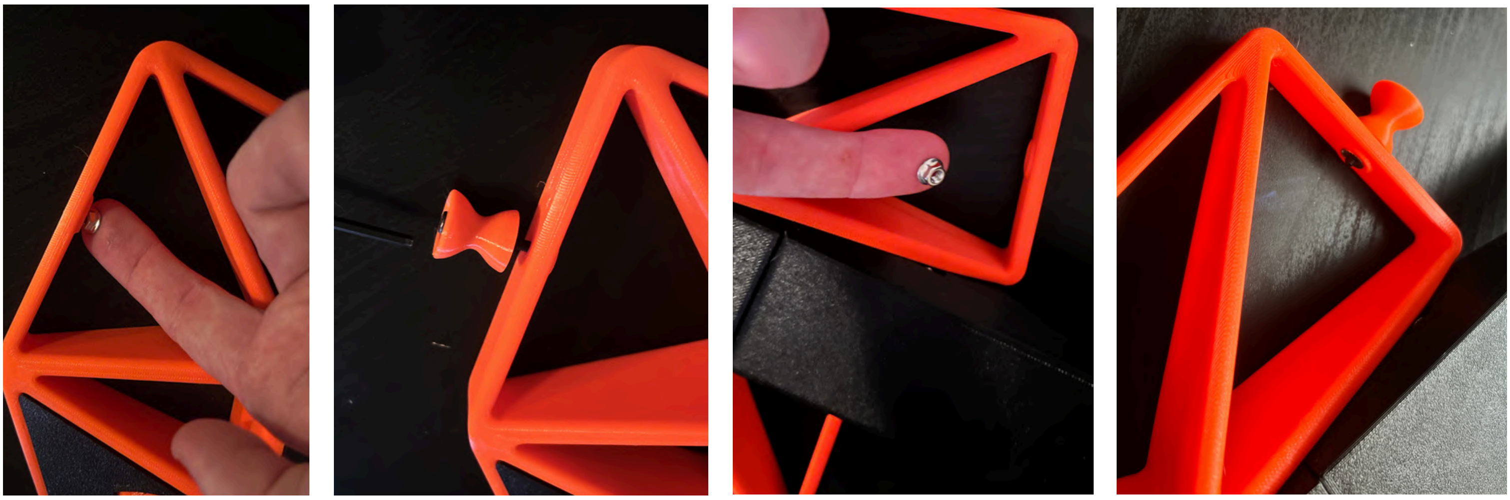
Parts Used

- (2) Strap pin
- (2) M3 Flange nuts
- (2) M3 Screws

Tools

- 2.5mm Hex wrench

Hold flange nuts in place and screw “strap pins” on



Admire your hard work

Step 12 Done

Step 13: Shred.

- Search: “How to play Clone Hero”
  or “How to play YARG”

- Take photo of your build and upload as a make on printables

https://www.printables.com/model/1828635-shredder-a-public-domain-guitar-rhythm-game-contro/comments

- Please leave any feedback. All is appreciated

THANK YOU SO MUCH!

HAPPY SHREDDING